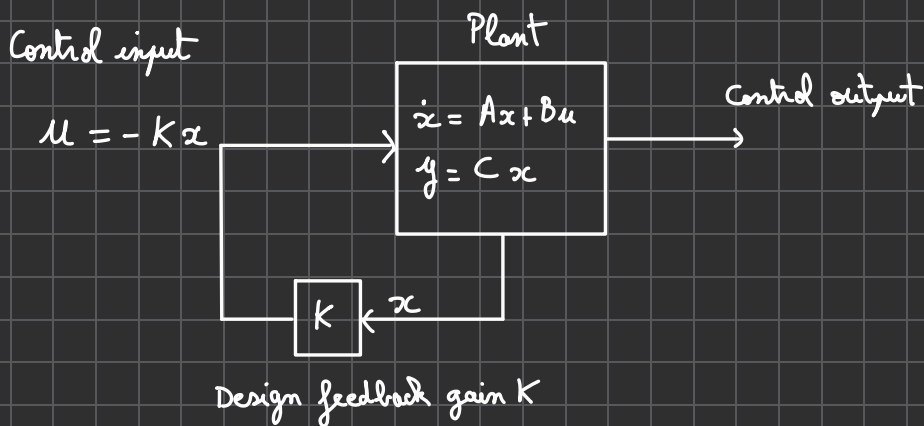




03) Feedback Controller design



$$\begin{aligned} \dot{x} &= Ax + Bu & u &= -Kx & \Rightarrow & \dot{x} = (A - BK)x \\ y &= Cx & & & & y = Cx \end{aligned}$$

system dynamics of feedback system is determined by $(A - BK)$.

Pole-placement method

$$\begin{aligned} \dot{x} &= (A - BK)x & \text{characteristic equation.} \\ y &= Cx & \Rightarrow \det(sI - A + BK) = 0 \end{aligned}$$

Eigenvalue of $(A - BK)$ = Solutions in characteristic equation = poles in closed loop system

By deciding feedback gain K , poles in closed-loop system $(A - BK)$ can be appropriately assigned.

Condition for pole-placement : The pair (A, B) is controllable.

(All poles of the closed-loop system should be located in the left half plane of complex plane)

Optimal regulator:

Find gain K , which minimize the following cost J

$$J = \int_0^{\infty} \boxed{x(t)^T Q x(t)} + \boxed{u(t)^T R u(t)} dt$$

Weight Matrices

Quickly converge to zero
if this term is small

Quick converge = Cost small

Input energy is small
if this term is small

Small Input Energy = Cost is small.

⇒ Optimal regulator problem achieve both quick response and minimum energy

→ Optimal gain is derived by following step:

① Solution P of Riccati Equation

$$A^T P + P A - P B R^{-1} B^T P + Q = 0$$

② Gain is obtained by using P

$$K = R^{-1} B^T P$$

Example of Weight function:

$$J = \int_0^{\infty} x(t)^T Q x(t) + u(t)^T R u(t) dt$$

$$Q = \begin{pmatrix} 100 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 100 \end{pmatrix}, R = 1$$

$$Q = \begin{pmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 10 \end{pmatrix}, R = 1$$

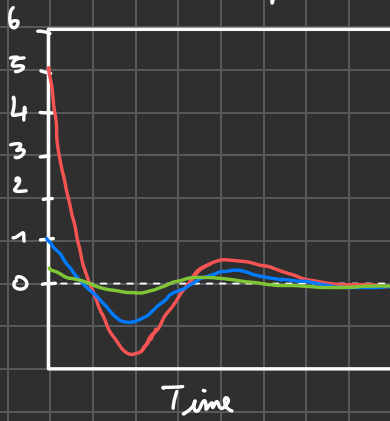
$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, R = 1$$

red when we want a quick response

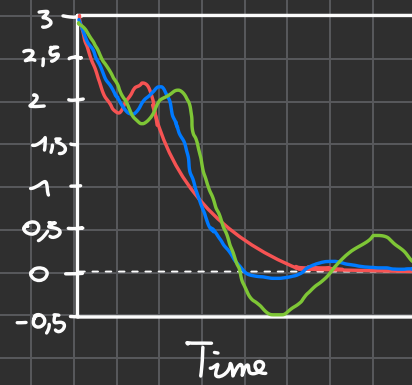
green when we want to minimize the input energy

Result of optimal control

Input response



Output response



(green is slow speed for converging to zero)